ASSIGNMENT

SUBMITTED BY:

MAMOONA TARIQ

SUBMITTED TO:

SIR ASAD MAJEED

CLASS:

BS_CS (A)

SUBJECT:

PROGRAMMING FUNDAMENTAL (LAB)

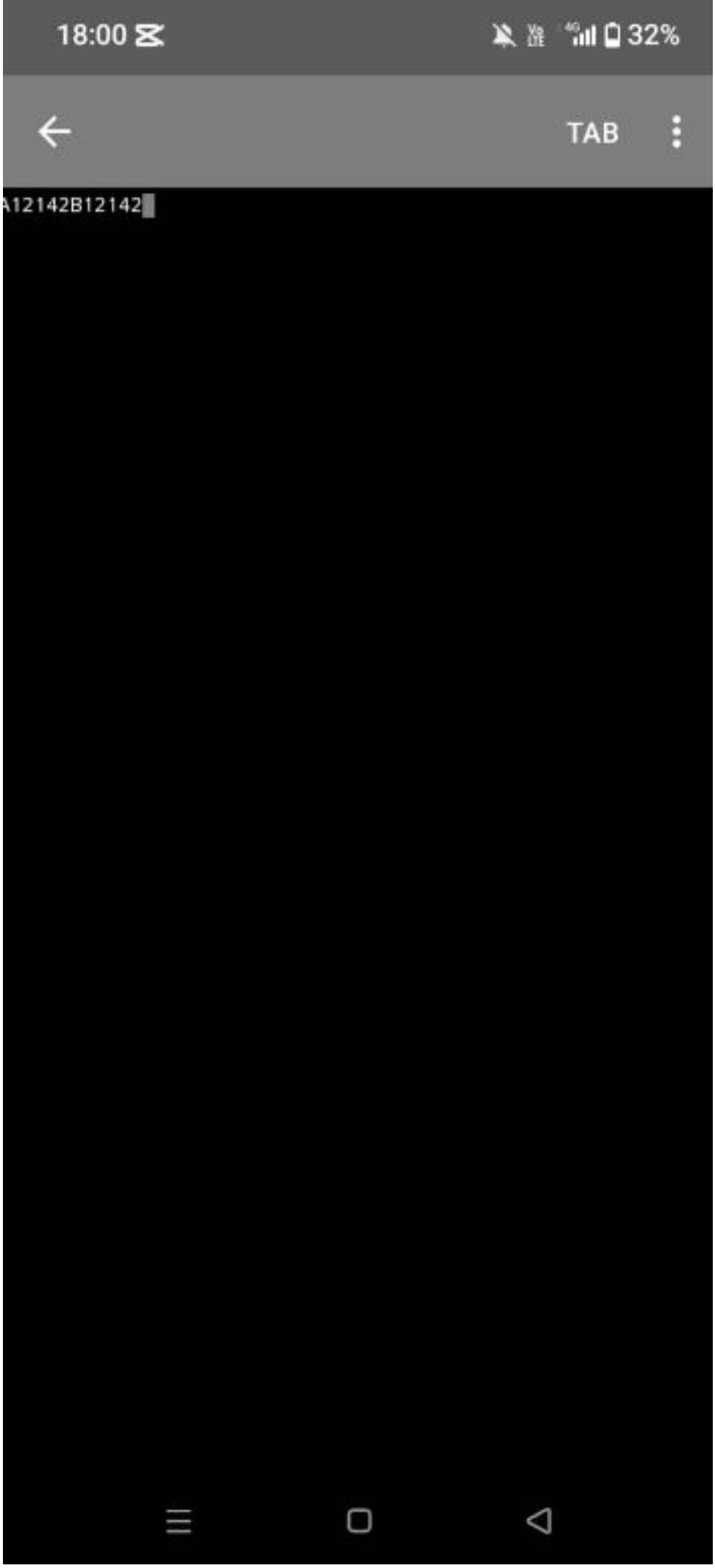
TOPIC:

LAB TASK

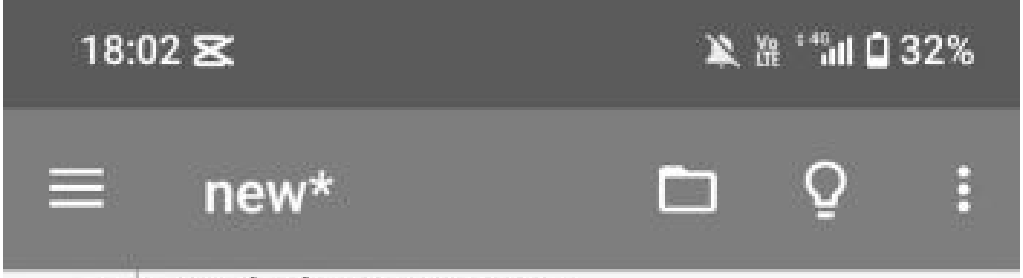
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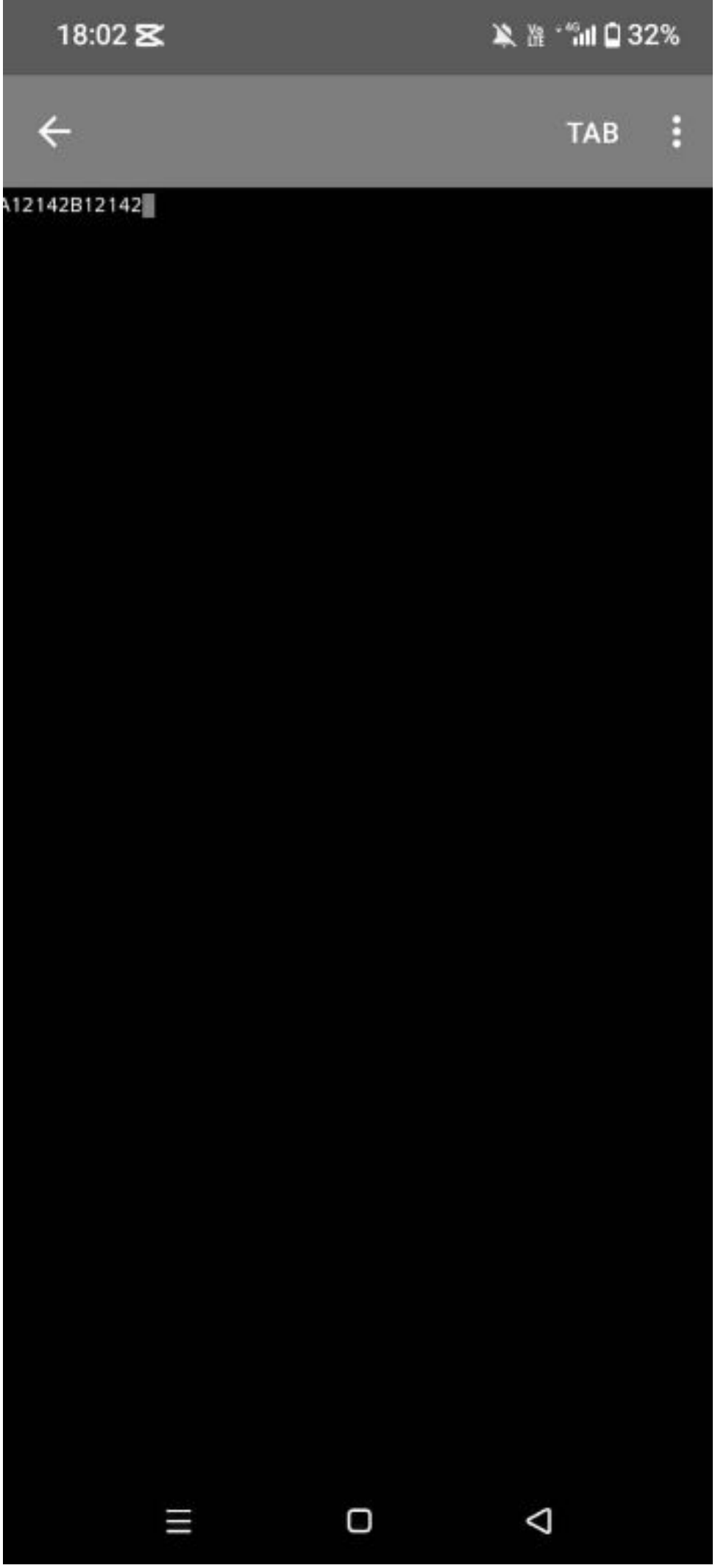


new*

```
1  #include<iostream>
2  using namespace std;
3  int main()
4  {
5     char letter;
6     letter=65;
7     cout<<letter<<'/n';
8     letter=66;
9     cout<<letter<<'/n';
10    cin>>letter;
11    return 0;
12 }
```



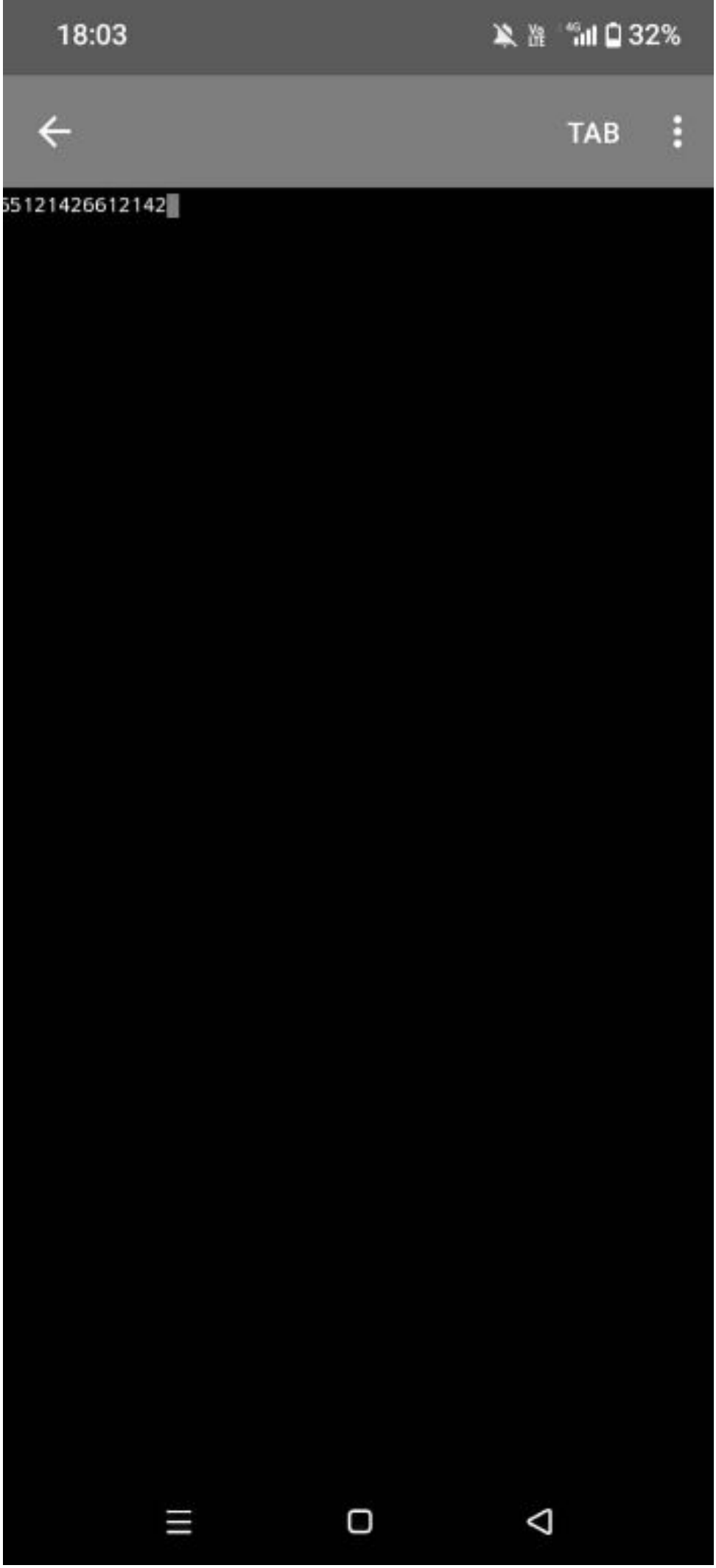
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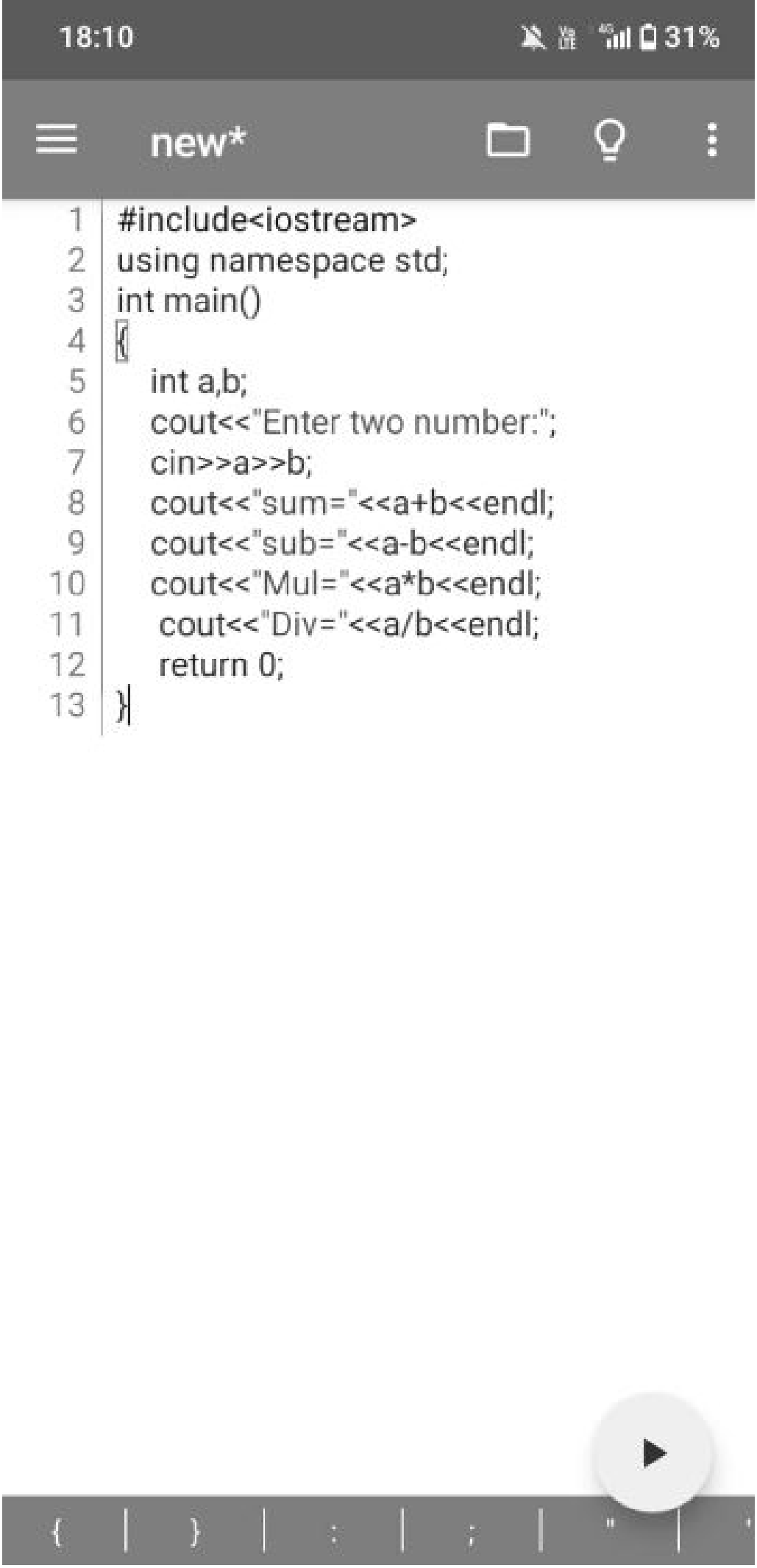
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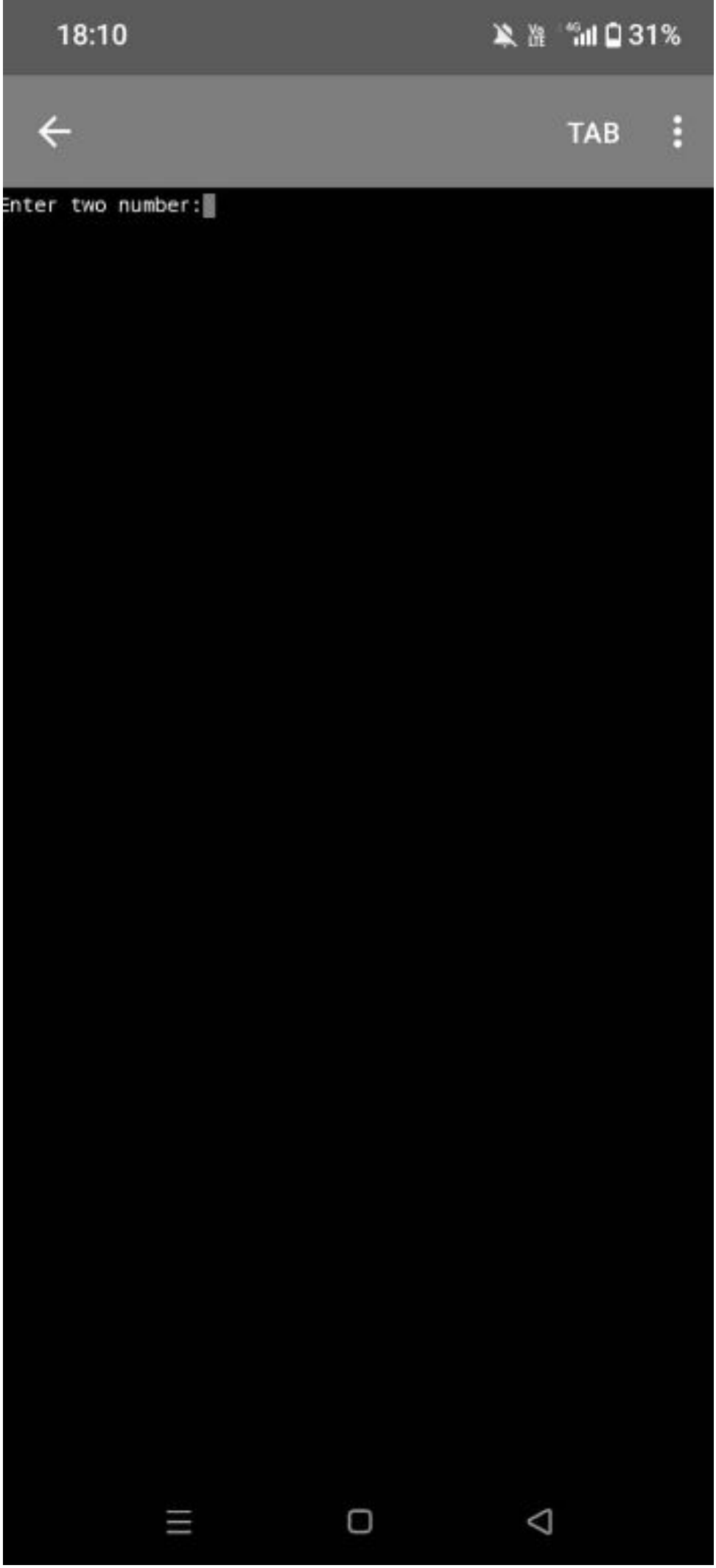
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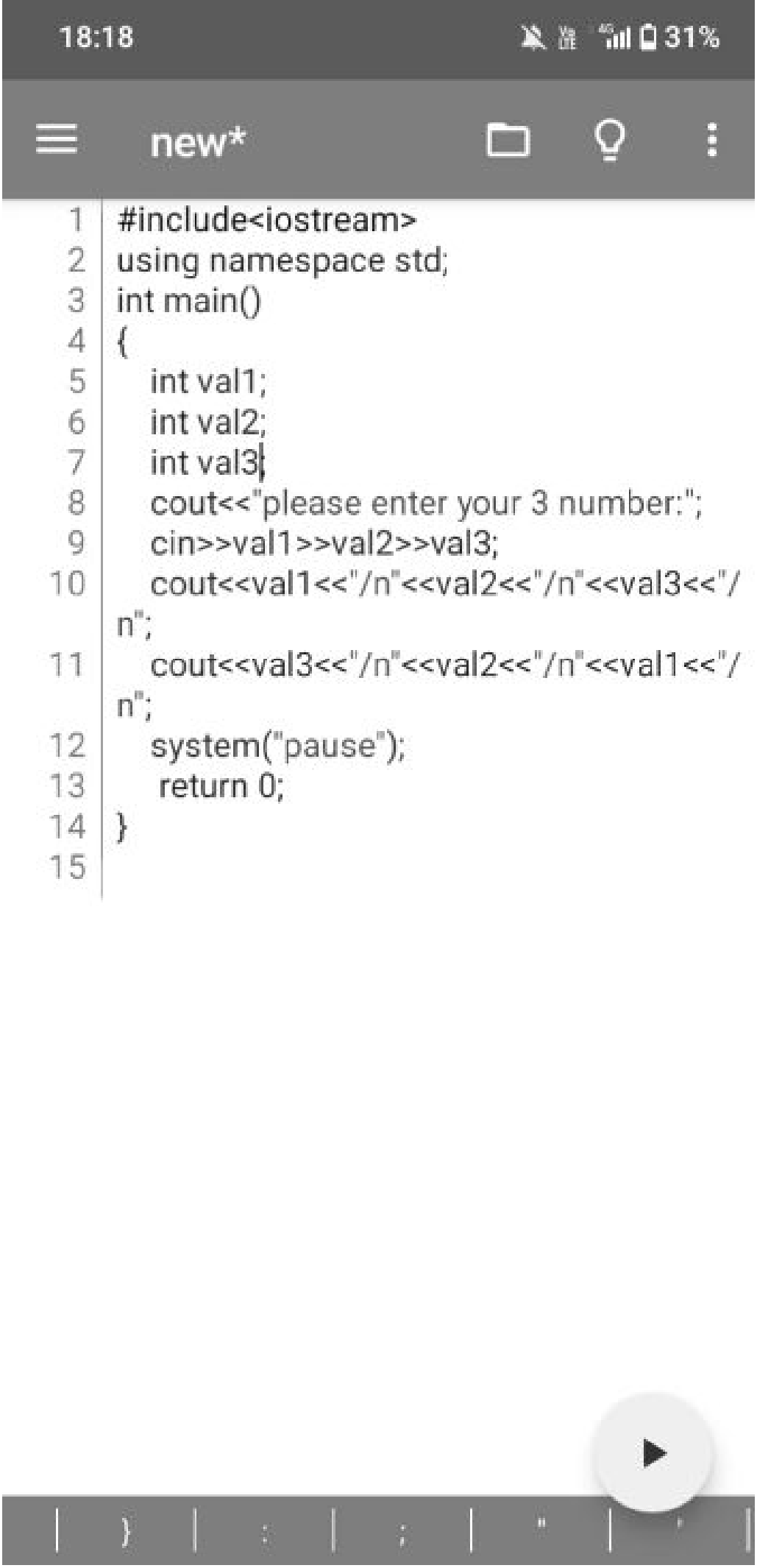
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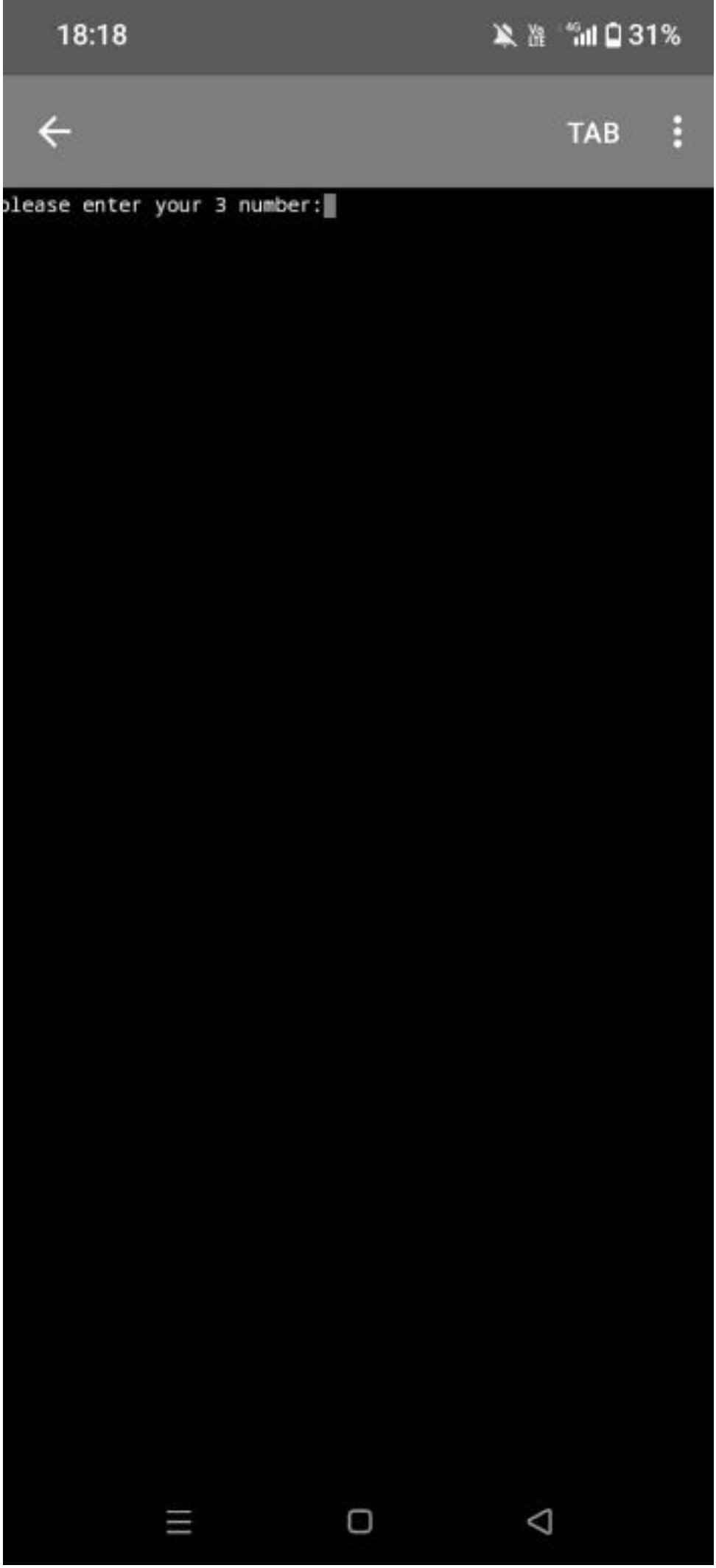
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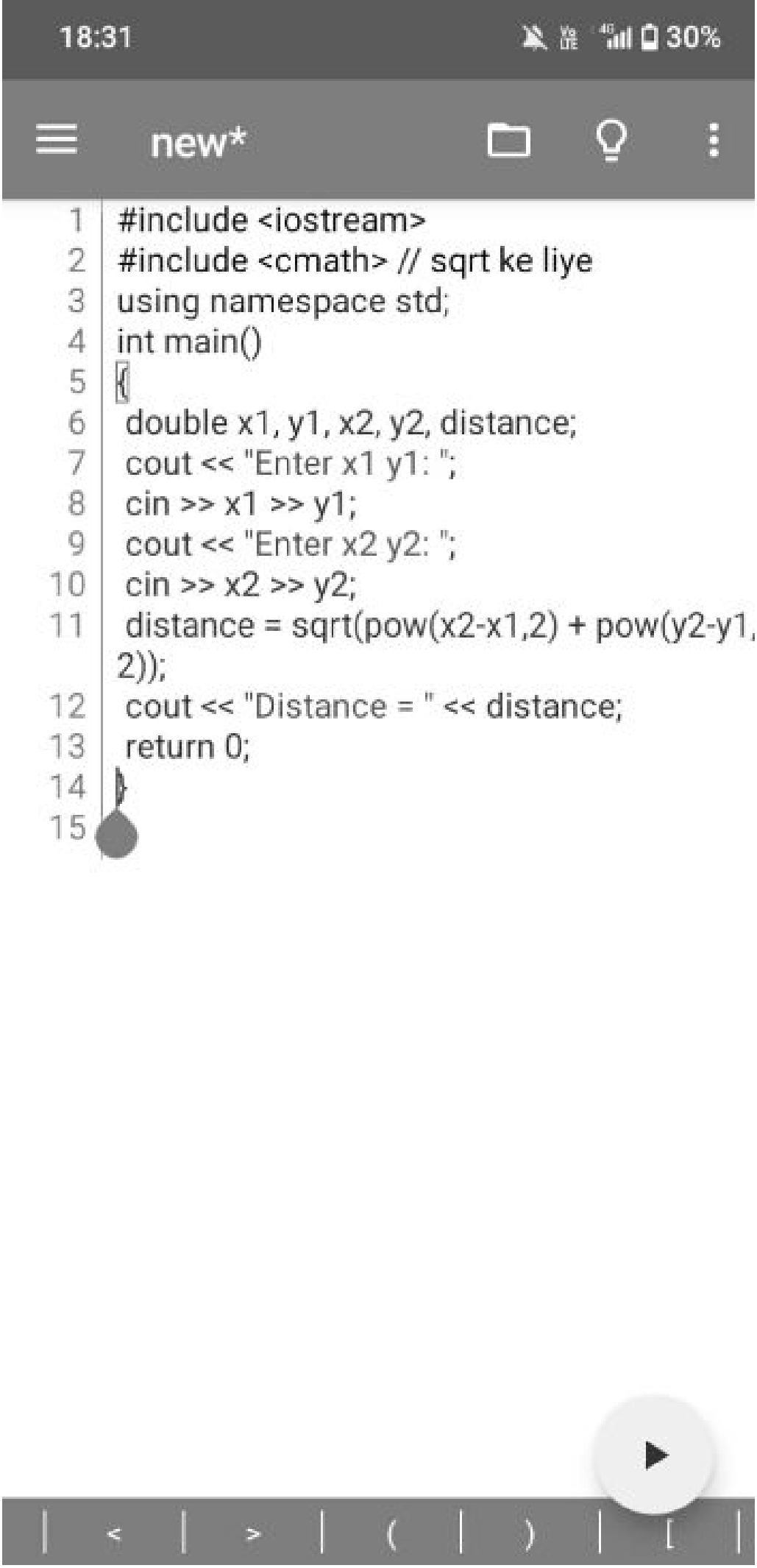
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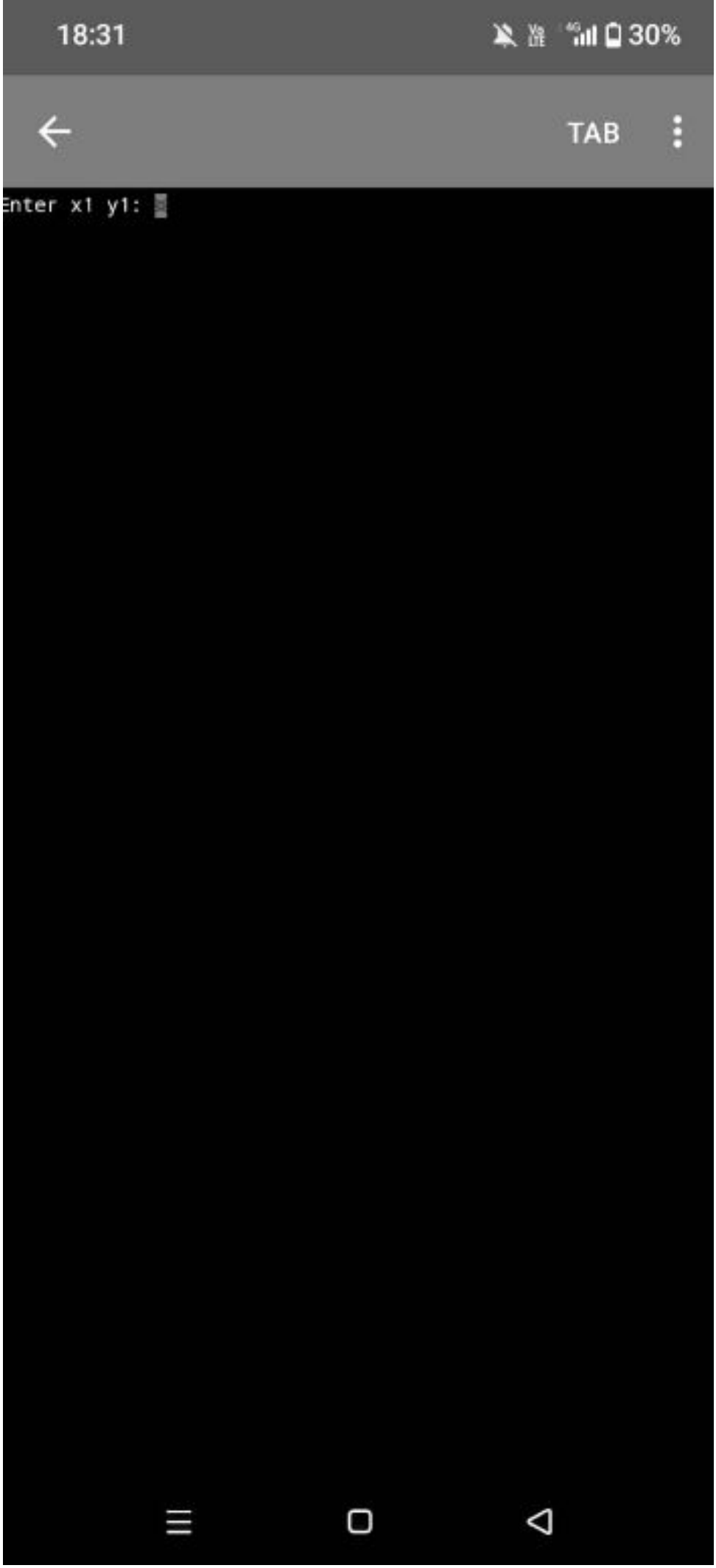
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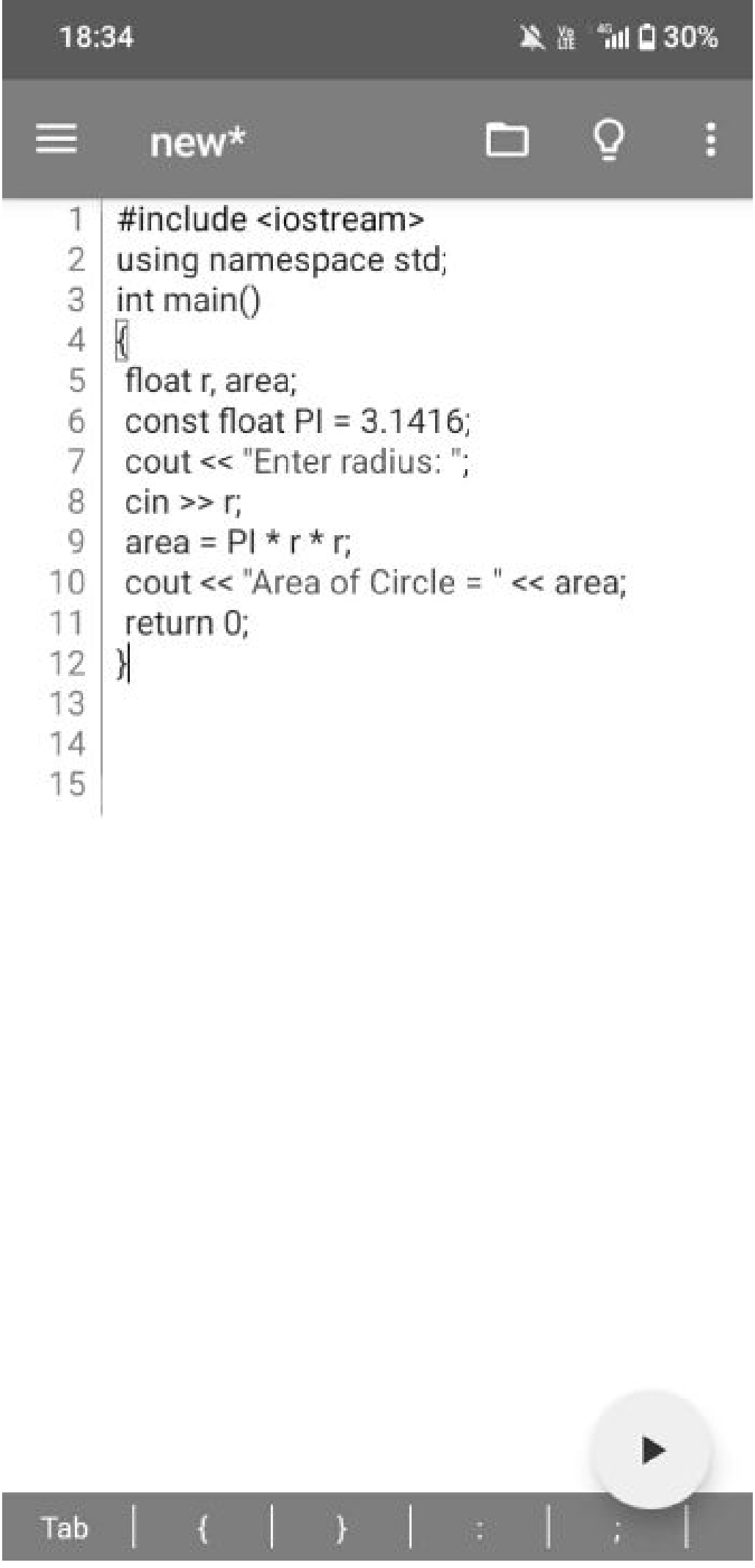
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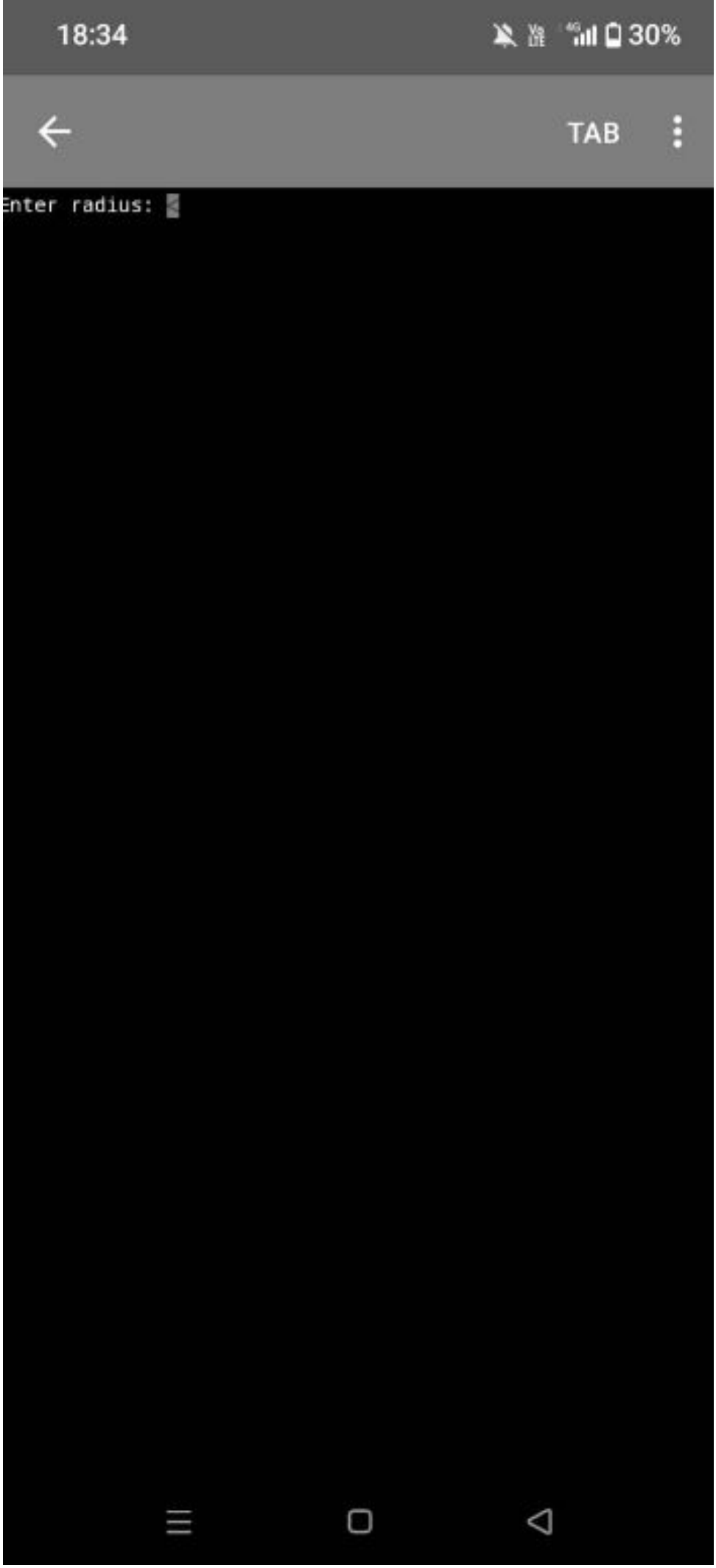
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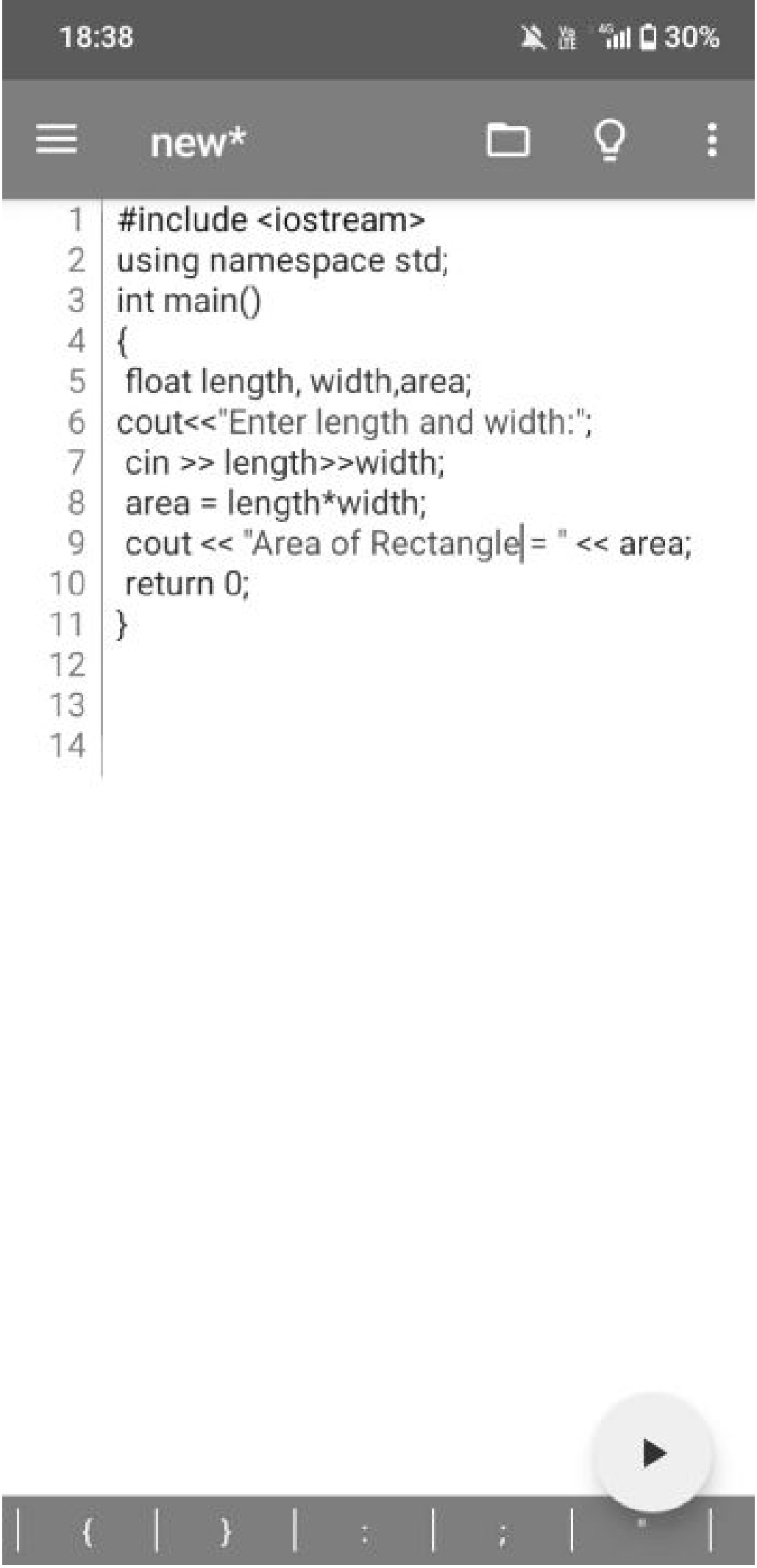
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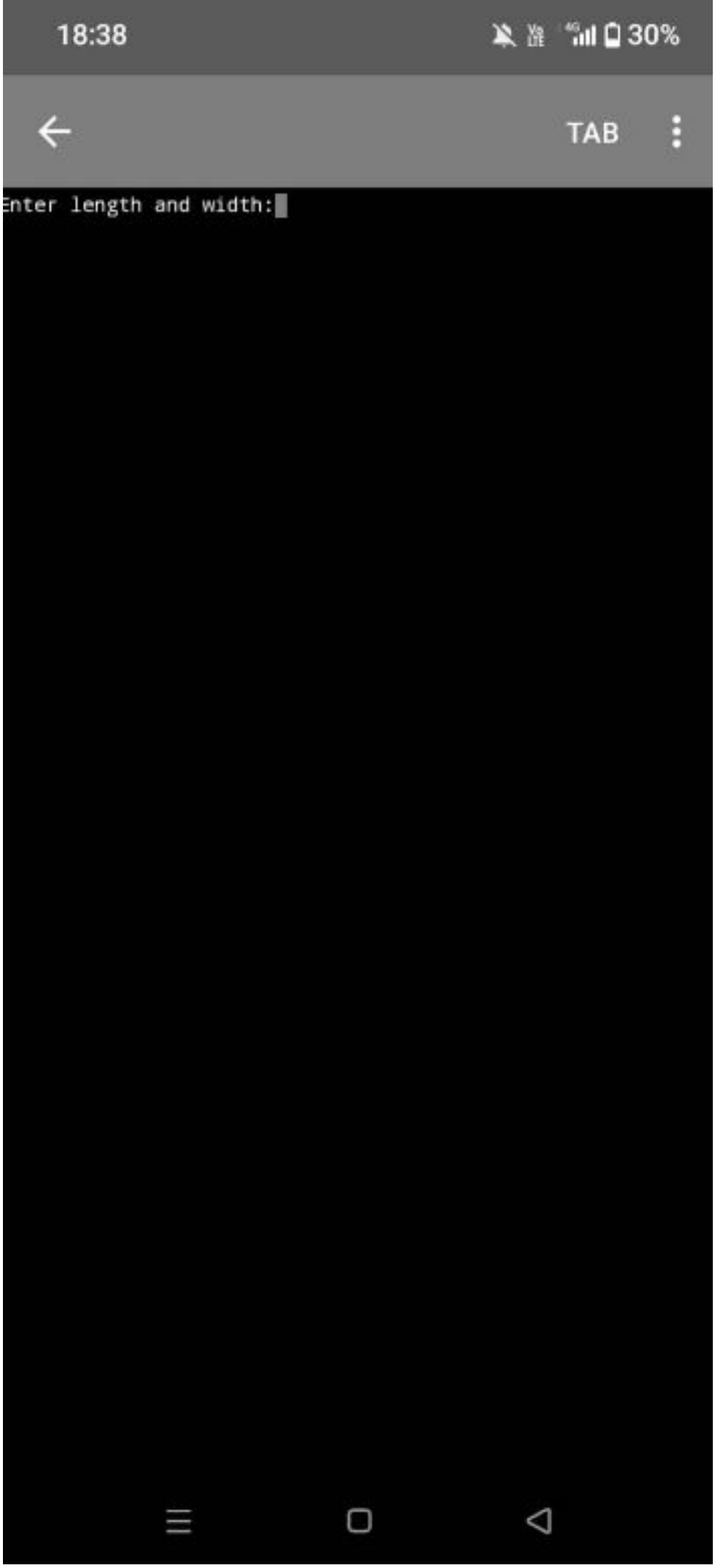
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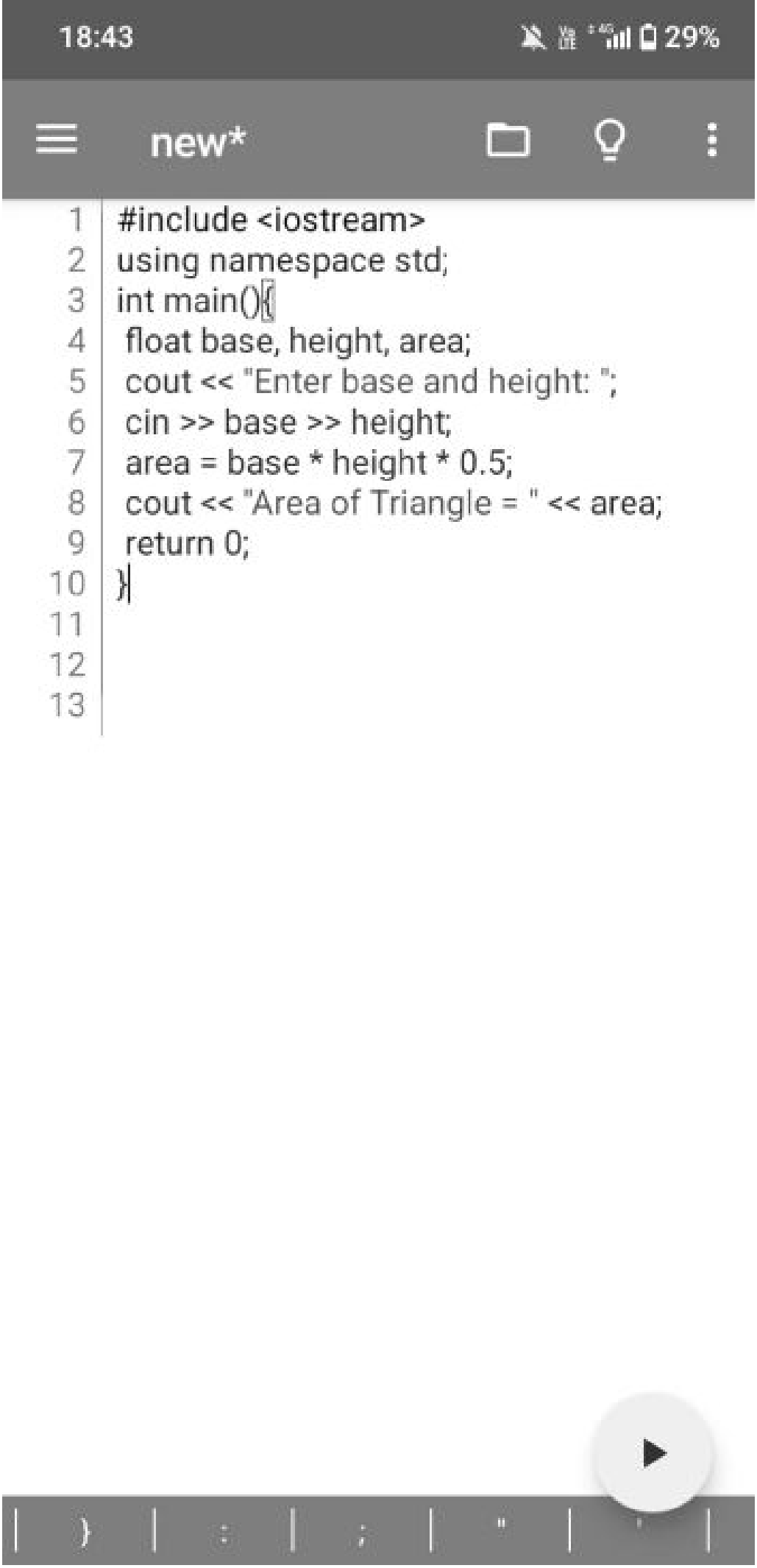
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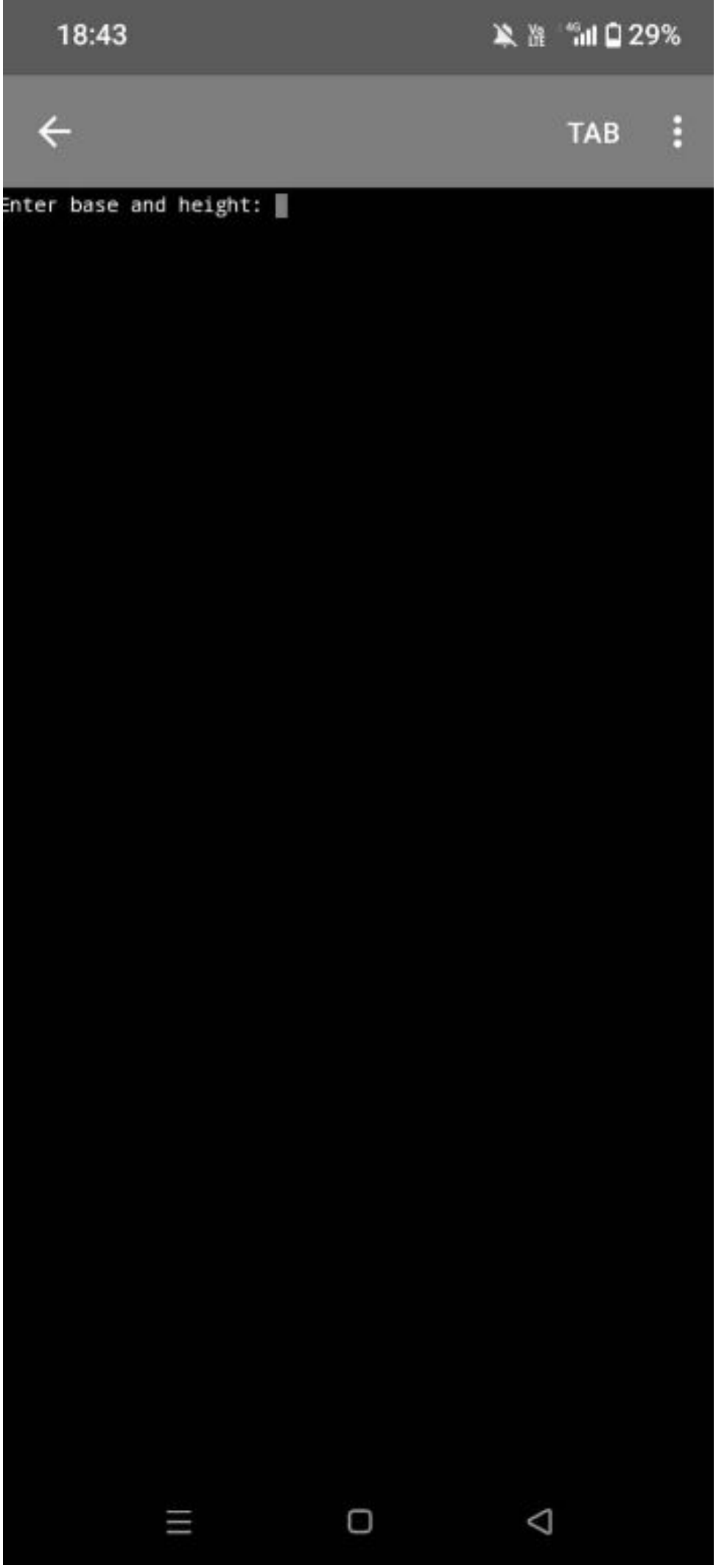
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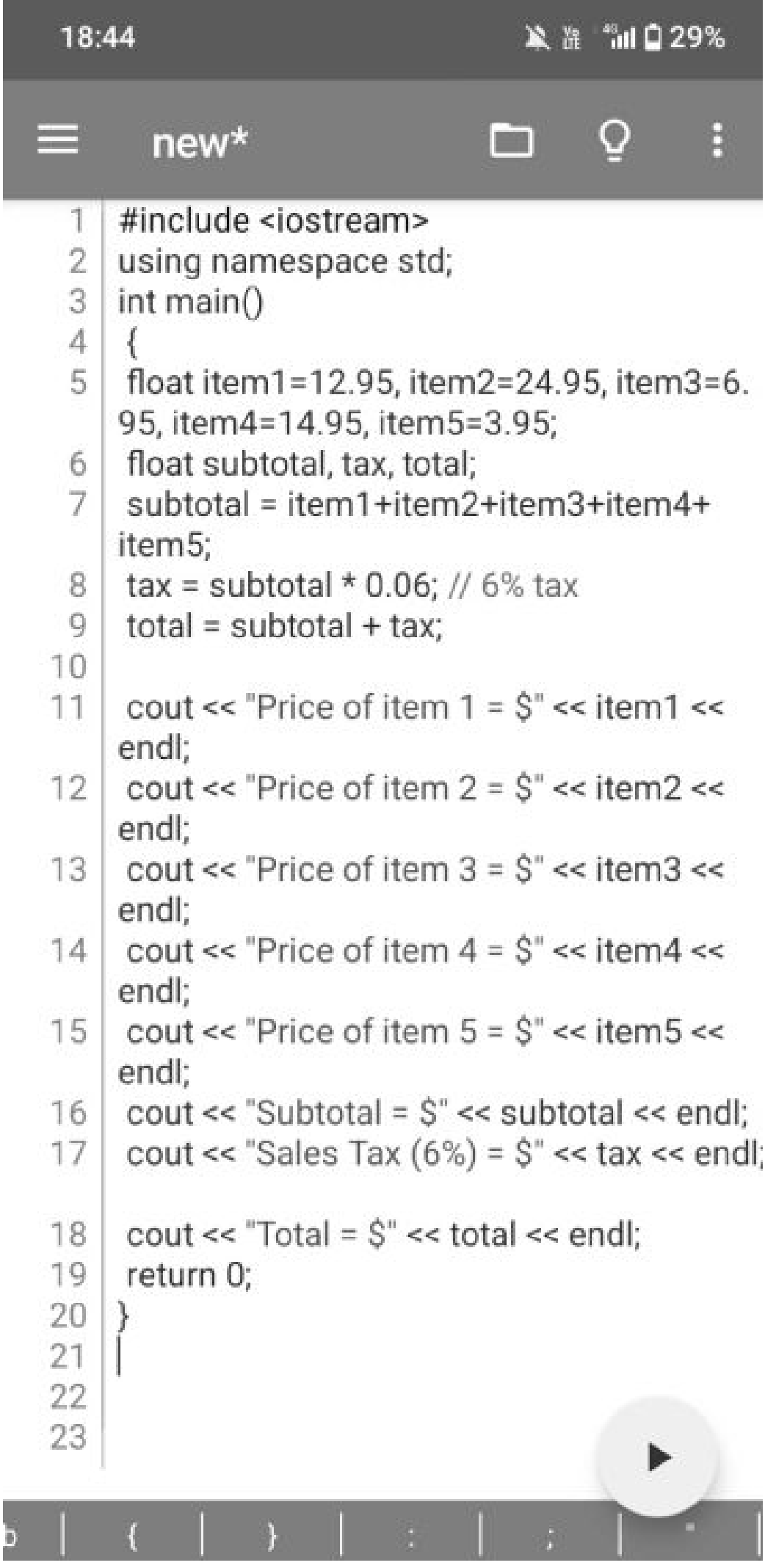
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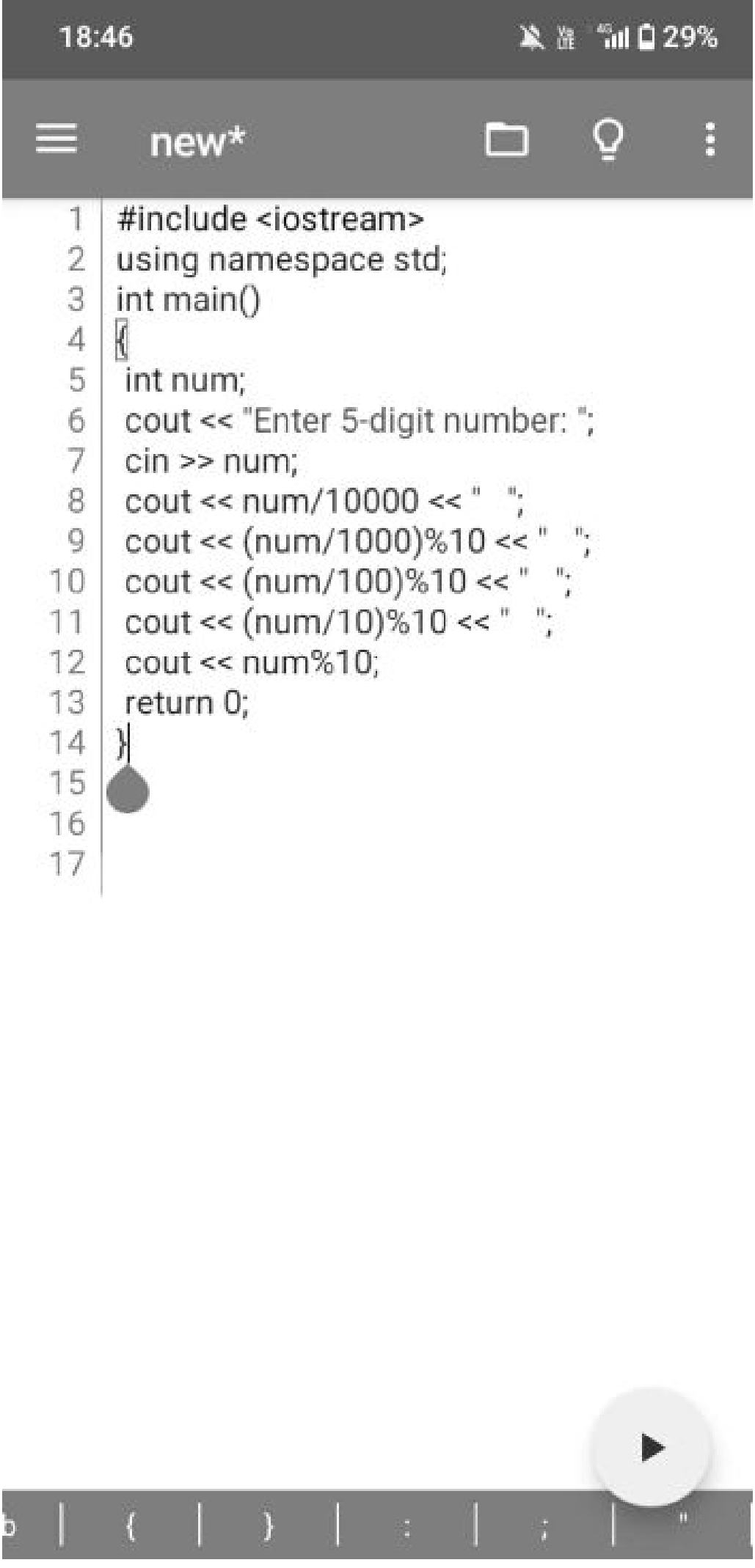
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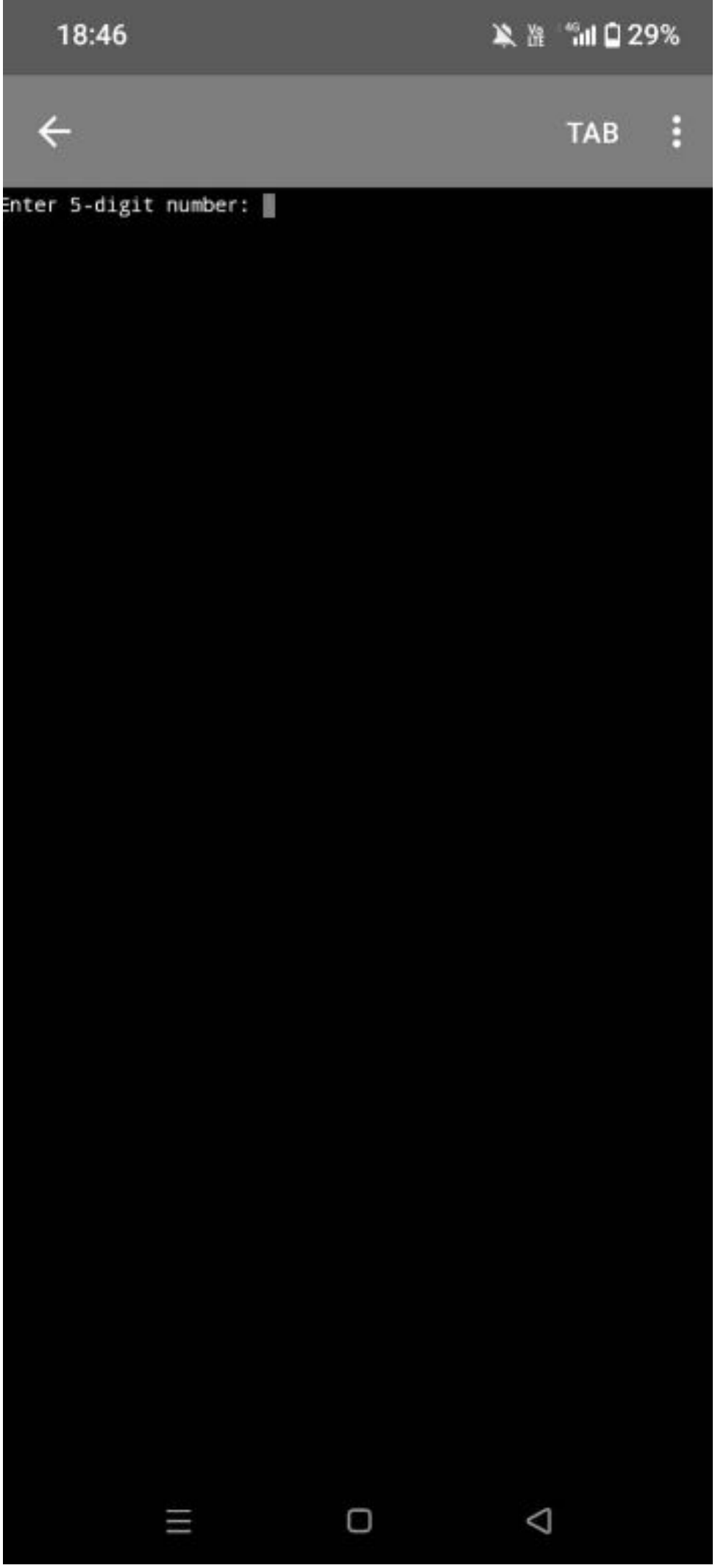
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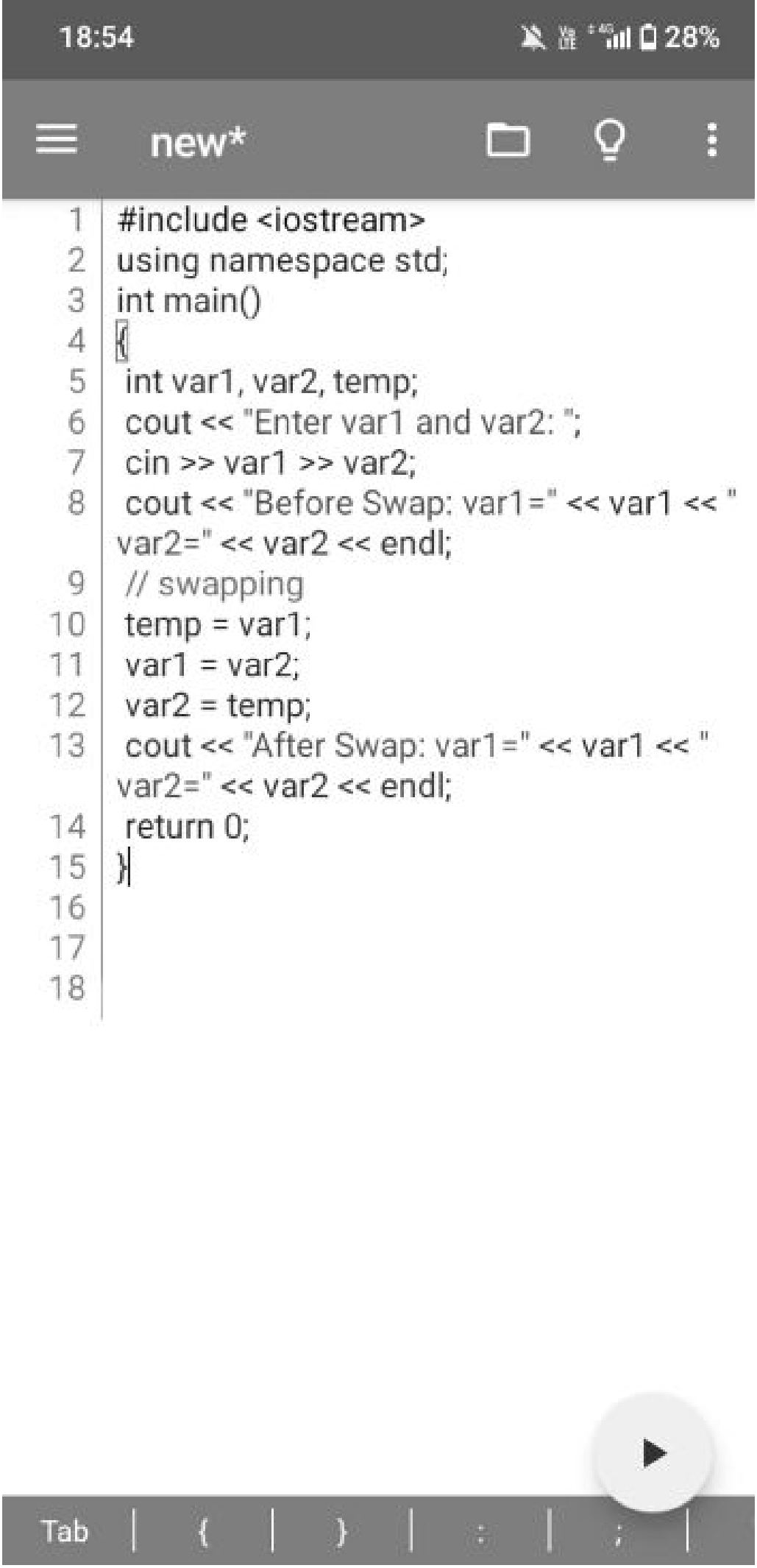
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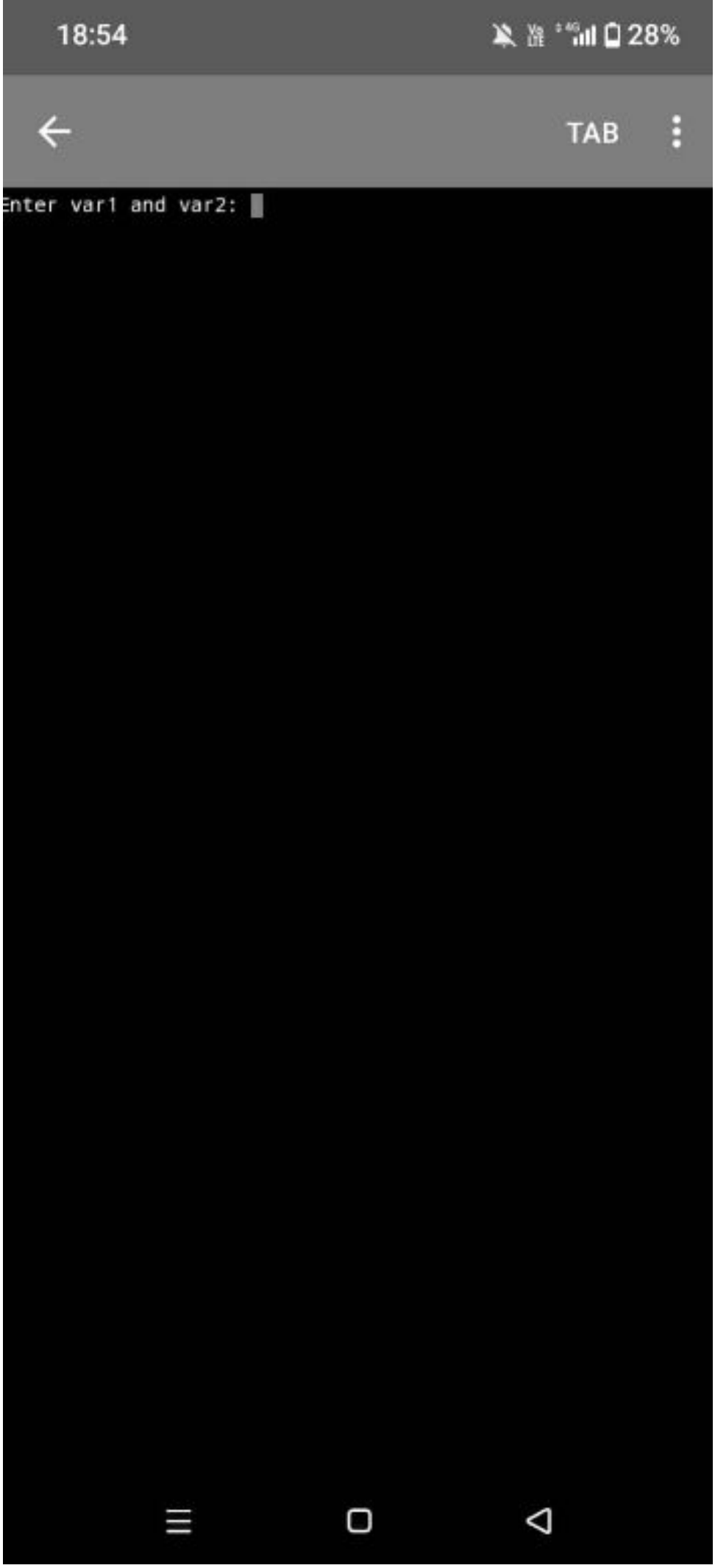
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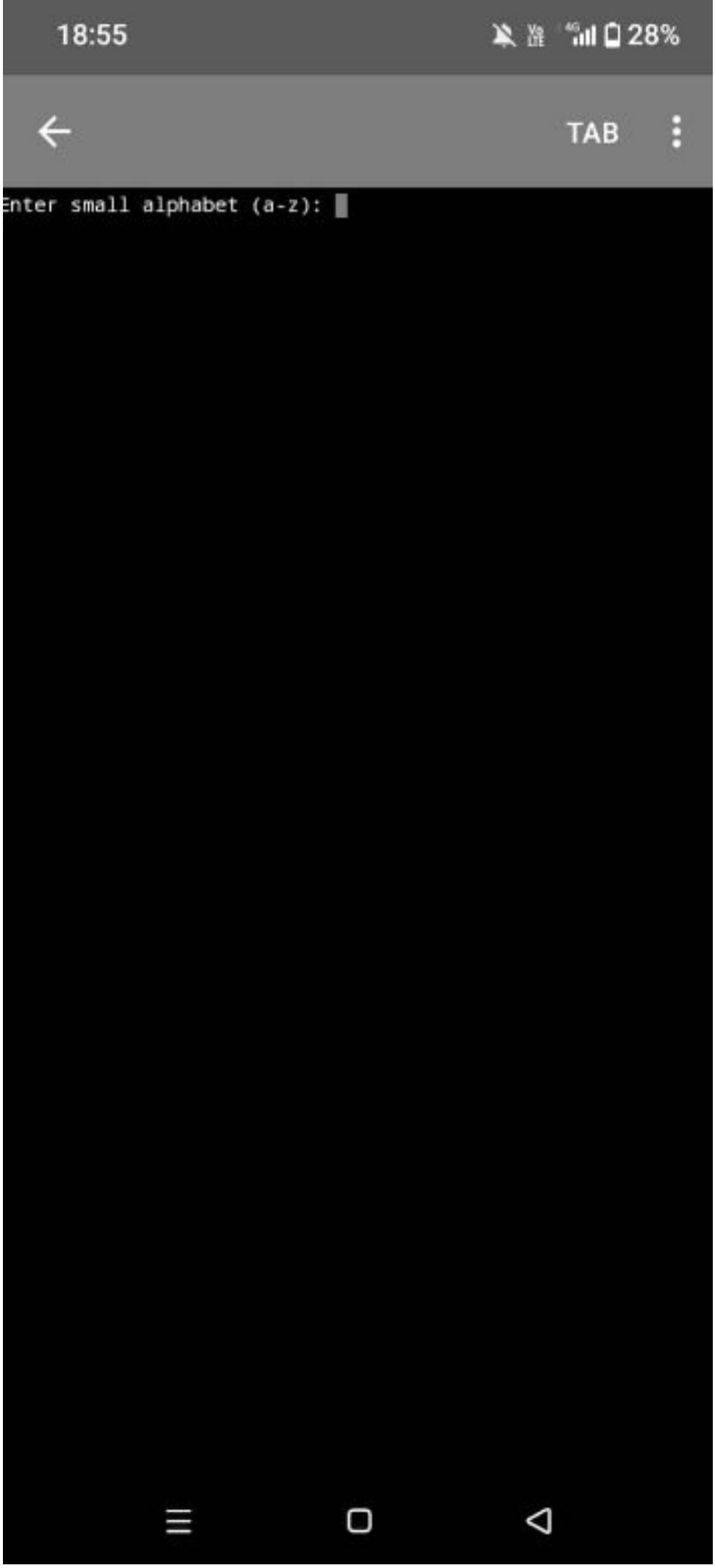
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new*

```
1  #include <iostream>
2  using namespace std;
3  int main()
4  {
5   char small;
6   cout << "Enter small alphabet (a-z): ";
7   cin >> small;
8   char capital = small - 32; // ASCII logic
9   cout << "Upper case = " << capital;
10  return 0;
11 }
12
13
14
```

Tab | { | } | : | ;

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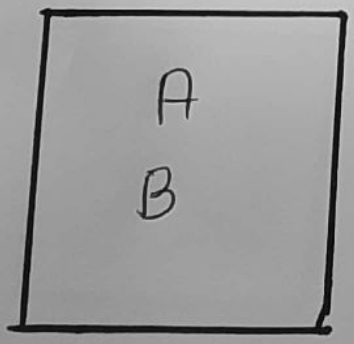
Task 1:

Run these programs and observe their output:

```
# include<iostream>
using namespace std;
 int main ()
{
 char = letter;
 letter = 'A';
cout<< letter<<'\n';
 letter = 'B';
cout<< letter<<'\n';
 cin>> letter;
 return 0;

}
```

output

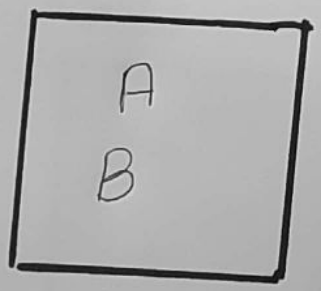


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Run these program and observe their output

```
#include<iostream>
using namespace std;
int main()
{
  char letter;
  letter = 65;
cout<< letter<<'\n';
  letter = 66;
cout<< letter<<'\n';
  cin>> letter;
  return 0;
}
```

output



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Run these program and observe their output.

```
# include <iostream>
using namespace std;
int main()
{
  int letter;
  letter = 'A';
cout<< letter<<'\n';
  letter = 'B';
cout<< letter<<'\n';
   cin>> letter;
   return 0;
  }
```

output

```
65
66
```

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Task 2:

Take two integers as input from the user apply athematic operations on them (+, -, *, //) as print them on Screen.

```
# include<iostream>
using namespace std;
int main()
{
  int a,b;
cout<<"Enter two numbers:";
 cin>>a>>b;
cout<<"Sum="<<a+b<<endl;
cout<<"Sub="<<a-b<<endl;
cout<<"Mul="<<a*b<<endl;
cout<<"Div="<<a/b<<endl;
  return 0;
}
```

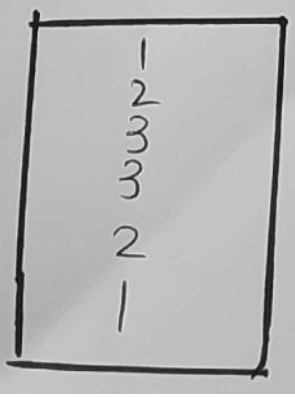
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Task3:

write the output of the following code.

```
#include "stdafx.h"
#include<iostream>
using namespace std;
int main()
{
 int val1;
 int val2;
 int val3;
cout<<"please enter your 3 numbers:";
cin >> val1>> val2>> val3;
cout<< val1<<"\n"<<val2<<"\n"<<val3<<"\n";
cout<<val3<<"\n"<< val2<<"\n"<<val1<<"\n";
 system("pause");
 return 0;
}
```

output



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Task4:

write a C++ program to calculate the distance between two points

Note: x_1, y_1, x_2, y_2 are all double values

Formula:

$$d = (x_2 - x_1)^2 + (y_2 - y_1)^2$$

```
#include<iostream>
#include<cmath> //sqrt ke liye
using namespace std;
  int main()
  {
   double x1, y1, x2, y2, distance;
   cout<<"Enter x1 y1 :";
   cin>>x1>>y1;
   cout<<"Enter x2 y2 :";
   cin>>x2>>y2;
   distance= sqrt(pow(x2-x1,2)+pow(y2-y1));
   cout<<"Distance = "<<distance;
     return 0;
   }
```

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Task 6:

(i) Calculate the Area of a circle (area = PI * r2)

```
#include <iostream>
using namespace std;
int main()
{
float r, area;
const float PI = 3.1416;
cout << "Enter radius:";
  cin >> r;
area = PI * r * r;
cout << "Area of circle=" << area;
    return 0;
  }
```

(ii) Area of Rectangle:

```
#include <iostream>
using namespace std;
  int main()
{
   float length, width, area;
  cout << "Enter Length and width:";
```

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```
    cin >> length >> width;
    area = length * width;
  cout<<"Area of Rectangle ="<<area;
   return 0;
  }
```

(iii) Area of Triangle (area = base-height)

```
#include<iostream>
using namespace std;
int main()
{
float base, height, area;
cout<<"Enter base and height:";
cin>> base >> height;
area= base * height * 0.5;
cout<<"Area of Triangle=" <<area;
return 0;
}
```

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Task 6:

```
#include<iostream>
using namespace std;
 int main()
{
  float item1=12.95 , item2=24.95, item3=6.
  item4=14.95 , item5=3.95;

 float Subtotal, tax, total;
Subtotal = item1+ item2 +item3 +item4 +item5;
  tax = Subtotal * 0.06; // 6% tax
   total = Subtotal +tax;
cout<<"Price of item 1 = $"<<item1<<endl;
cout<<"price of item2 = $"<<item2<<endl;
cout<<"price of item3 = $"<< item3 <<endl;
cout<<"price of item4 = $"<< item4<<endl;
cout<<"price of item5 = $"<<item5<<endl;
cout<<"Subtotal = $"<< Subtotal <<endl;
cout<<"Sales Tax (6%) = $"<<tax<<endl;
 cout <<"Total = $"<< total << endl;
  return 0;
}
```

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Bouns Task : Break Number.

```
#include <iostream>
using namespace std;
 int main ()
{
  int num;
cout<< "Enter 5- digit number:";
  cin >> num;
 cout<<num/10000<<" ";
cout<<(num/10000)% 10<<" ";
 cout<<(num/100)% 10<<" ";
cout<<(num/10) % 10<<" ";
  cout<< num % 10;
  return 0;
  }
```

Home Task1

```
# include<iostream>
using namespace std;
 int main ()
 {
```

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```
int var1 , var2 , temp;
cout<<"Enter var1 and var2:";
 cin>> var1 >> var2;
cout<<"Before swap: var1="<<var1<<"var2="<<var2
 <<endl;
 // swapping
  temp=var1;
  var1=var2;
  var2 = temp;
 cout<<"After swap: var1="<<var1<<"var2="<<
  var2<< endl;
   return 0;
  }
```

Task2:

```
#include<iostream>
using namespace std;
 int main()
 {
  char small;
cout<<"Enter small alphabet (a-z):";
 cin>> small;
 char   capital = small - 32;
```

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```
cout<<"Upper case = "<< capital;
    return 0;
}
```

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